ME EN 7210 Optimal Controls **Spring 2026**
Assignment #6 **Due Thursday, March 26 at the beginning of class**

This assignment MUST be typeset in either LaTex or MS Word. Any hand-written work will not be accepted and if submitted, the entire assignment will receive a zero grade! Please start on this assignment early and come see Dr. Leang with any questions.

Problem 1 Find the optimal feedback control for the plant

$$\dot{x}_1(t) = x_2(t)$$
$$\dot{x}_2(t) = -2x_1(t) + 4x_2(t) + 5u(t)$$

with performance criterion

$$J = \frac{1}{2} f_{11} \left[x_1(t_f) - 4\right]^2 + \frac{1}{2} f_{22} \left[x_2(t_f) - 2\right]^2 + \frac{1}{2} \int_0^{t_f} \left[5x_1^2(t) + 2x_2^2(t) + 4u^2(t)\right] dt$$

and initial conditions as $\mathbf{x}(0) = [1 \quad 2]'$ and the final state $\mathbf{x}(t_f)$ is free, where t_f is specified.

Assume that the final time is 5. Be sure to plot the Riccati coefficients, optimal states, and optimal input as a function of time.

Problem 2: Optimal Control of a Motor System

Consider the Quanser motor system shown in Fig. 1 below (System from the Small Robotics Lab). Note, this system should be familiar to you from other courses, such as ME EN 5200/6200 or 5210/6210.

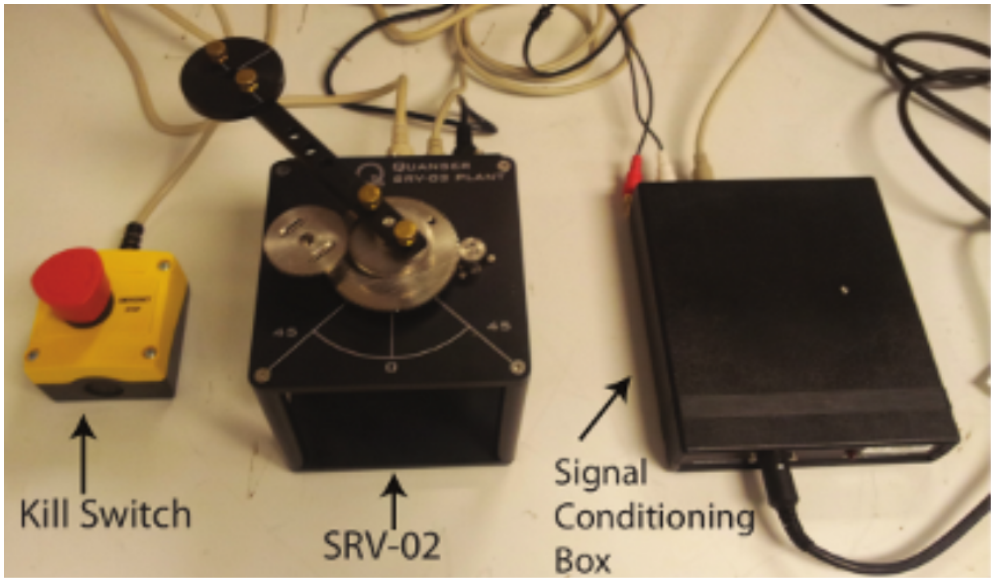

Figure 1. The system has a motor/mass system (SRV-02) connected to a computer via an interface board (NI-6024).

The system is typically configured as a position feedback control system and can be modeled by the following block diagram:

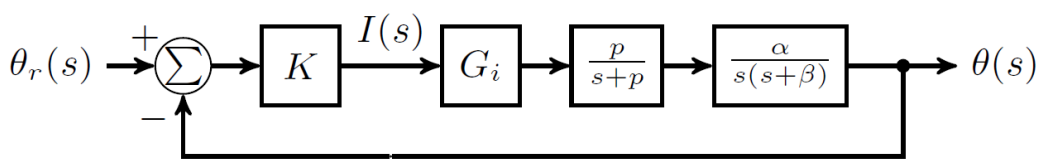

Figure 2. Closed-loop system block diagram.

In Fig. 2, $G_i = 10$ is the drive gain, $p = 20$ is the filter parameter, $\alpha = 7{:}5$ and $\beta = 1{:}67$ are constants found experimentally, I is the commanded current input, θ is the rotational position, and θ_r is the desired (or reference) rotational position. Note that the transfer function $\alpha/(s(s+\beta)$ is the motor transfer function. Note that the open-loop model contains G_i, the filter, and the transfer function of the motor.

(a) Before doing any design work on the controller, your first task is look at the dynamics of the motor system in simulation. Do to this, you will need to apply a series of inputs I(t) and simulate the response, then plot the simulated response using Matlab. Show the details of this step, along with plots to justify the overall behavior of the open-loop system model. Explain what you see and whether the simulated response makes sense in terms of how it models the real physical system.

(b) Design a finite-time full-state LQR controller for the motor-mass system. It should minimize the energy (e.g., control effort and energy in the states). It should start from rest at angle 0° and move to 45°. Let the final time be 1 second.

Define the weighting matrices, F, Q, and R.
Try the following cases:

$Q = 0$ and $F \neq 0$
$F = 0$ and $Q \neq 0$
$F \neq 0$ and $Q \neq 0$

Plot the results, including the gains vs. time, response vs. time, and input vs. time.